probability which the equations can, however, still describe. And the final image of the fifth lesson is transformed thus:



Our universe may have been born from a bounce in a prior phase, passing through an intermediate phase in which there was neither space nor time.

Physics opens windows through which we see far into the distance. What we see does not cease to astonish us. We realize that we are full of prejudices and that our intuitive image of the world is partial, parochial, inadequate. The Earth is not flat, it is not stationary. The world continues to change before our eyes as we gradually see it more extensively and more clearly. If we try to put together what we have learnt in the twentieth century about the physical world, the clues point towards something profoundly different from our instinctive understanding of matter, space and time. Loop quantum gravity is an attempt to decipher these clues, and to look a little further into the distance.